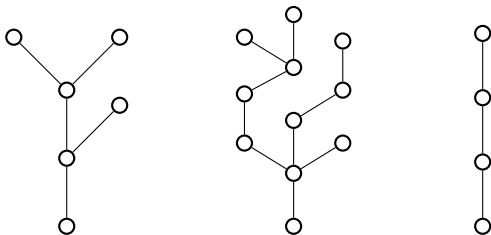


Building a Graph, One Edge at a Time

Aurora Hiveley
Adv. Kristin Heysse

Department of Mathematics, Statistics, and Computer Science
Macalester College



Friday Night Plans



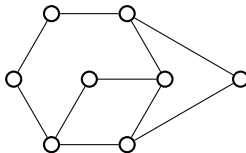
Rules

- 1.) No plans? $\Rightarrow$ Make plans!
- 2.) Both have plans? $\Rightarrow$ No plans made!

Graph = collection of vertices with edges between them

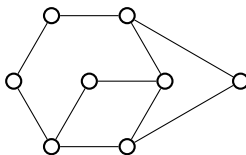
Definitions

Graph = collection of vertices with edges between them



Definitions

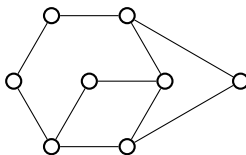
Graph = collection of vertices with edges between them



Permutation = random ordering of objects in a set

Definitions

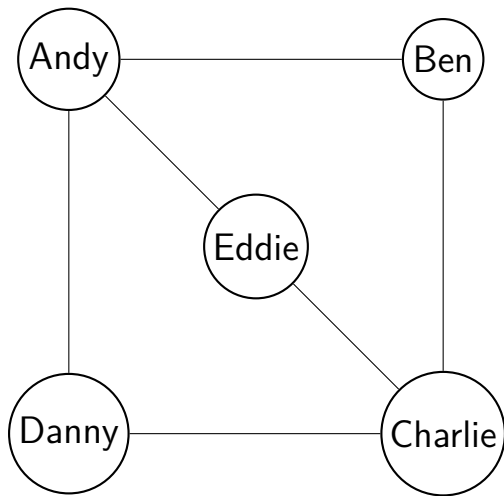
Graph = collection of vertices with edges between them



Permutation = random ordering of objects in a set

$$\{1, 2, 3, 4\} \rightarrow 2341 \text{ or } 4132 \text{ or } 1243$$

Graphs

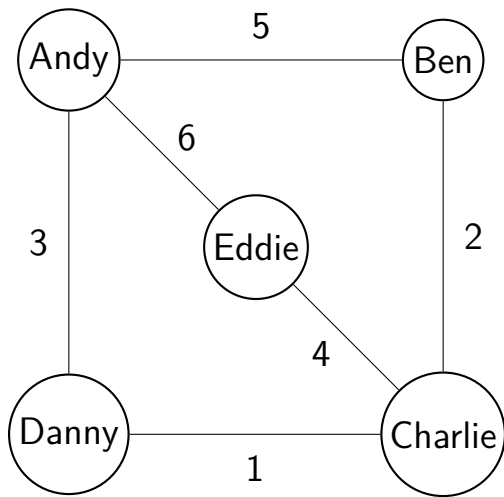


Vertices = Friends

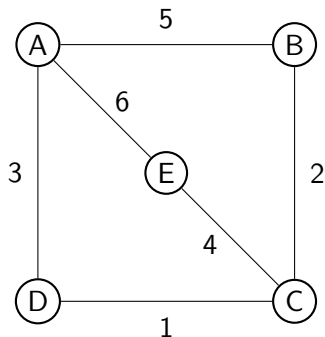
&

Edges = Contacts

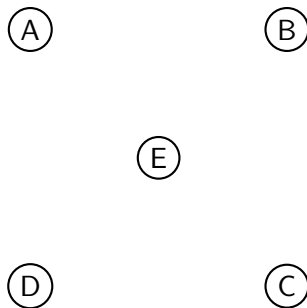
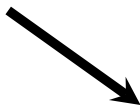
Graphs



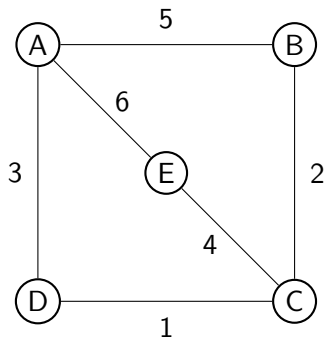
Reconstruction



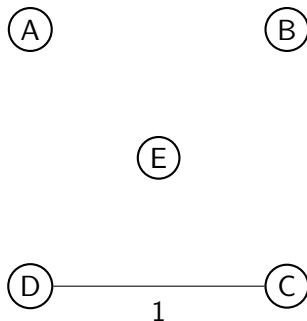
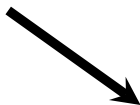
- 1.) No plans? $\Rightarrow$ Make plans!
- 2.) Both have plans? $\Rightarrow$ No plans made!



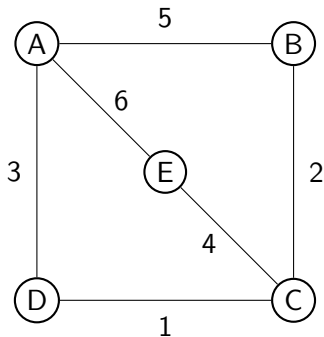
Reconstruction



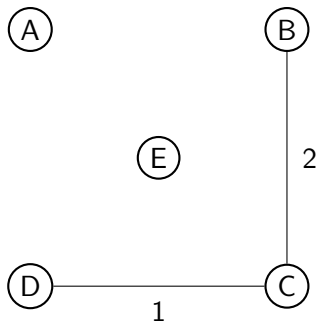
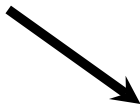
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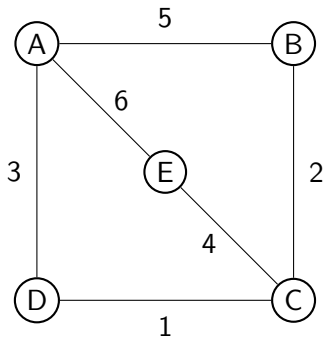
Reconstruction



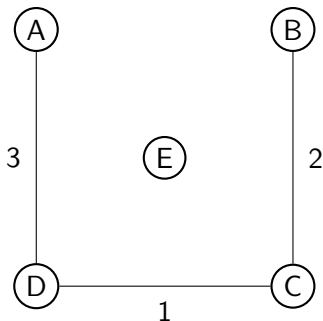
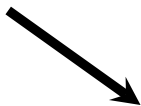
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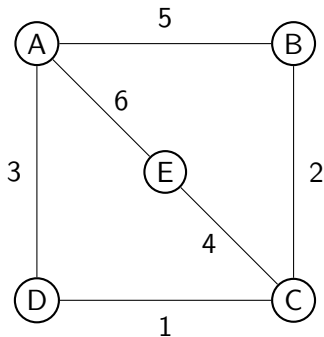
Reconstruction



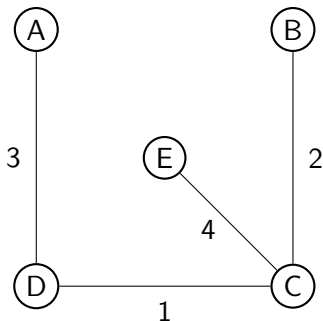
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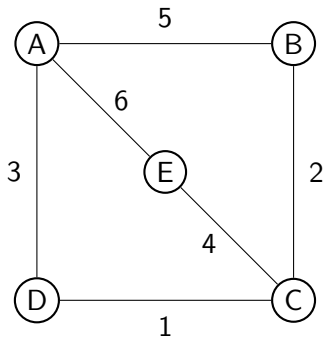
Reconstruction



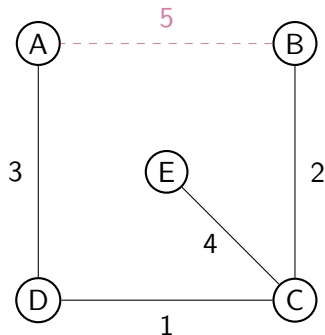
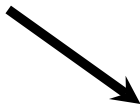
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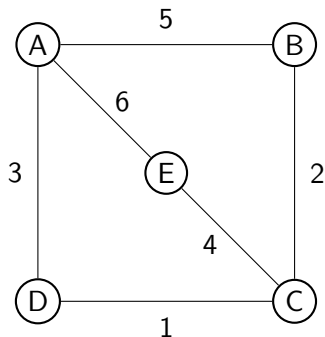
Reconstruction



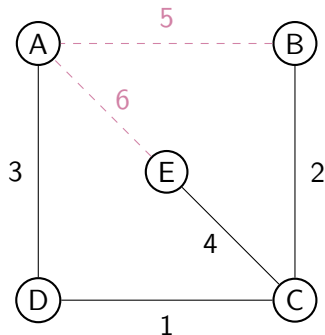
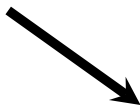
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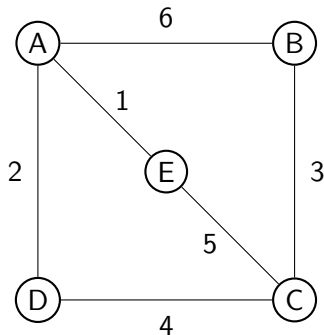
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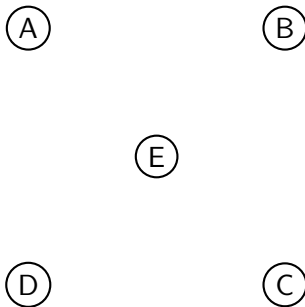
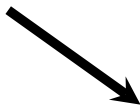
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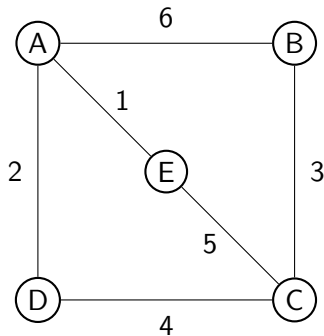
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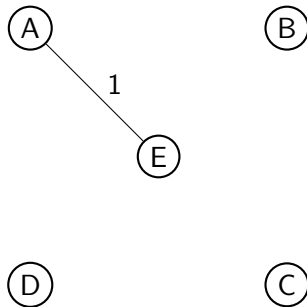
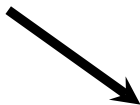
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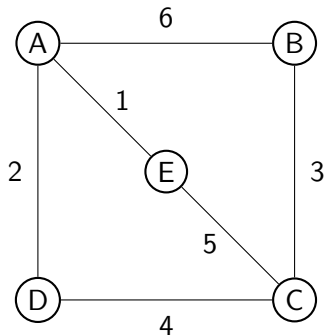
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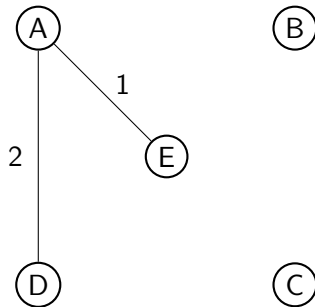
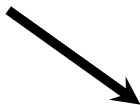
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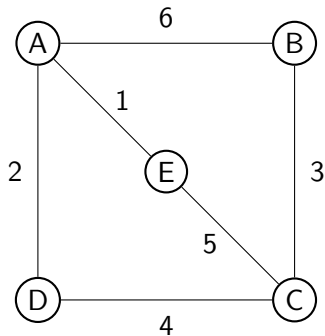
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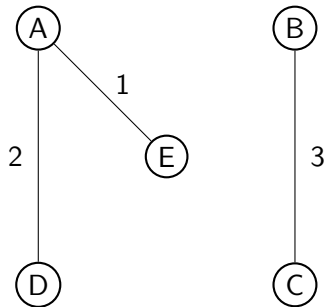
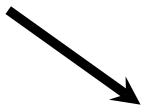
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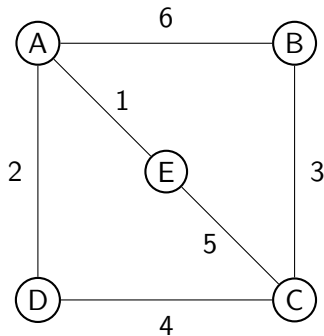
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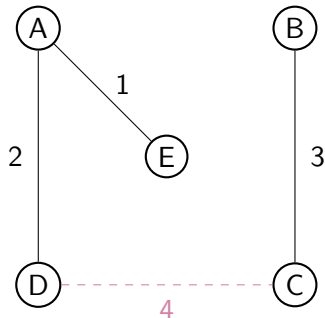
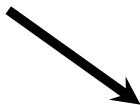
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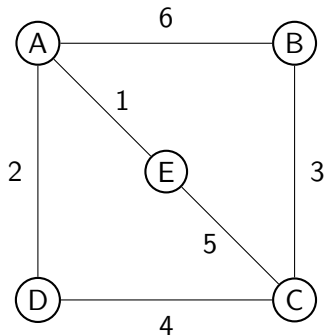
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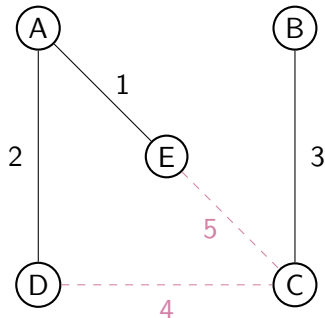
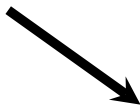
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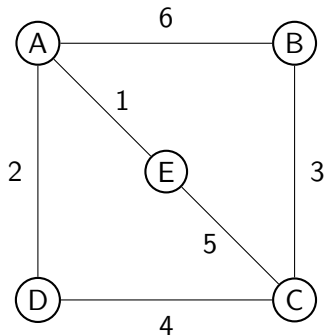
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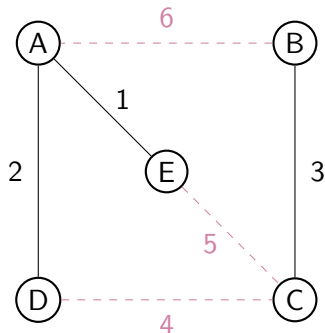
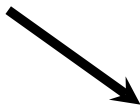
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Reconstruction

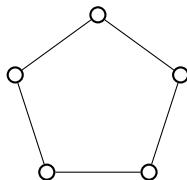
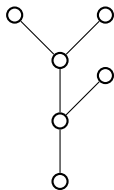


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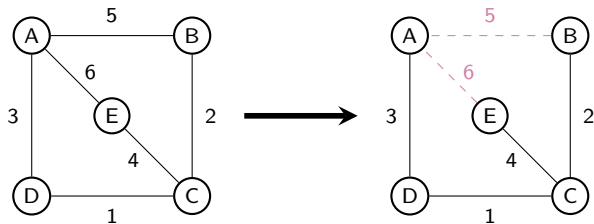
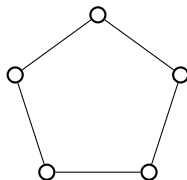
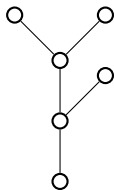
Tree = Graph without cycles

Tree = Graph without cycles



Trees

Tree = Graph without cycles

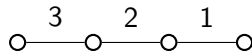
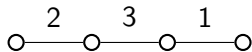
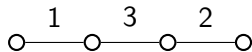
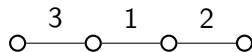
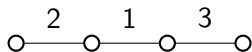
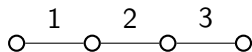


Paths



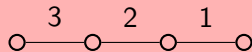
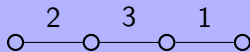
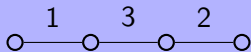
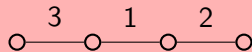
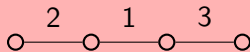
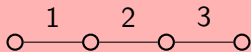
Paths

P_4 

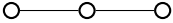


Paths

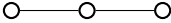
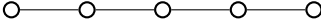
P_4 



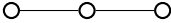
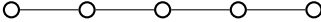
1-Component Counts

Path	Graph	$\#(P_n, k = 1)$
P_2		
P_3		
P_4		
P_5		
P_6		

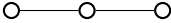
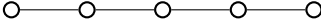
1-Component Counts

Path	Graph	$\#(P_n, k = 1)$
P_2		1
P_3		2
P_4		4
P_5		8
P_6		16

1-Component Counts

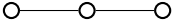
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		$\Downarrow$
		2^{n-2}

1-Component Counts

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$$\#(P_n, k = 1) = 2^{n-2}$$

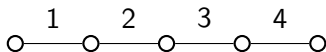
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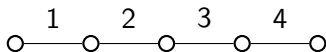
$$\#(P_n, k = 1) = 2^{n-2}$$

... why?

Proof for P_5



Proof for P_5



1234

2134

2314

2341

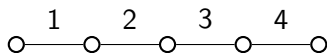
3214

3241

3421

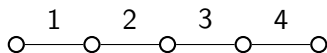
4321

Proof for P_5

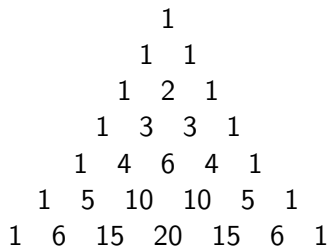


1234	}	1
2134	}	3
2314		
2341		
3214	}	3
3241		
3421		
4321	}	1

Proof for P_5



1234	}	1
2134	}	3
2314		
2341		
3214	}	3
3241		
3421		
4321	}	1



What else?

$$\#(P_n, k = 2) = 2^{n-1}(2^n - (n + 1))$$

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$\hookrightarrow$ Why????

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$$\#(P_n, k = 2) = 2^{n-1}(2^n - (n + 1))$$

$\hookrightarrow$ Why????

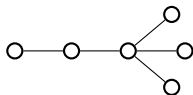
Other graph families:

What else?

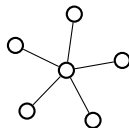
$$\#(P_n, k=2) = 2^{n-1}(2^n - (n+1))$$

$\hookrightarrow$ Why????

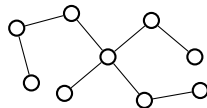
Other graph families:



Brooms



Stars



Spiders

Questions?

Related courses: MATH 279, 375, 379

